BRAC University (Department of Computer Science and Engineering)

CSE 221 (Algorithms) for Summer 2024 Semester

Assignment 1

Full Marks: 60

Deadline : Wednesday, 10 July, 2024, 11:59pm

DEADLINE WILL NOT BE EXTENDED

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1. Explain the time complexity of the following code snippets in regards of the Big-O notation: **[5x3 = 15]**

a.

```
for (i=0; i<n; i+=5) {
        for (j=1; j<n; j*=3) {
                for (k=0; k<30; k++) {
                        arr[i] = arr[i] +1;
                }
                int t = 0;
                for (m=n; m>1; m/=2) {
                        a = b + c;
                }
        }
}
```

b.

```
for i in range (1, n):
        j= 1
        while j < i*i:
                j= j – 1
```

c.

```
for(k = n; k > 1; k = k/n){
        for (j = 1; j*j <= n; j = j+ 2){
                for (i = 1; z < = n; i = i*4){
                        z = z+i
                }
        }
        for (m = 1; m*m < n; m = m*4){
                b = b + 1
        }
}
```

2. Express the following running time T(n) with an asymptotic bound. Any method is acceptable as long as you show calculations. **[10x3 = 30]**

   a. $T(n) = 8T(n/4) + n\sqrt{n}$

   b. $T(n) = T(n - 3) + n$

   c. $T(n) = 625T(n/5) + n^3$

3. Propose a **divide and conquer** algorithm to compute **the square of an n digit number**. Take Karatsuba's algorithm for multiplication as inspiration. Use your analytical knowledge as much as possible. **[6+4+5 = 15]**

   a. Express your solution by a **flowchart**.

   b. Convert the flowchart in **code**.

   c. **Analyze the complexity** of your code in regards of Big-O notation.

Note: Name your **handwritten scanned pdf file** with your student_ID. Suppose your student_ID is 12345678, then the file name will be **12345678_assignment_1.pdf.**
Failure to follow submission guidelines will be severely penalized.